

5G FUNDAMENTALS AND ARCHITECTURES
(EC23712)

ASSIGNMENT 2

TEAM 4

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Question: Configure two Network Slices with different S-NSSAI values, provision subscriber slice information in MongoDB, and attach a UERANSIM UE requesting a specific slice.

Objective:

The objective of this assignment is to demonstrate end-to-end 5G Network Slicing on an open-source virtualized 5G mobile network architecture. By configuring distinct Single Network Slice Selection Assistance Information (S-NSSAI) values—comprising Slice/Service Type (SST) and Slice Differentiator (SD)—within the Open5GS subscriber database (MongoDB) and core configuration, the task validates that a simulated User Equipment (UE) using UERANSIM can request, authenticate, and successfully establish a PDU session bound to a designated network slice.

Configuration:

5G Network Slicing allows a single physical infrastructure to be logically partitioned into multiple isolated virtual networks. Each slice is identified by its S-NSSAI ($S-NSSAI = SST + SD$):

Target Network Slice (Requested):

- SST (Slice/Service Type): *1* (Standard eMBB - Enhanced Mobile Broadband)
- SD (Slice Differentiator): *000002*
- Data Network Name (DNN/APN): *internet*

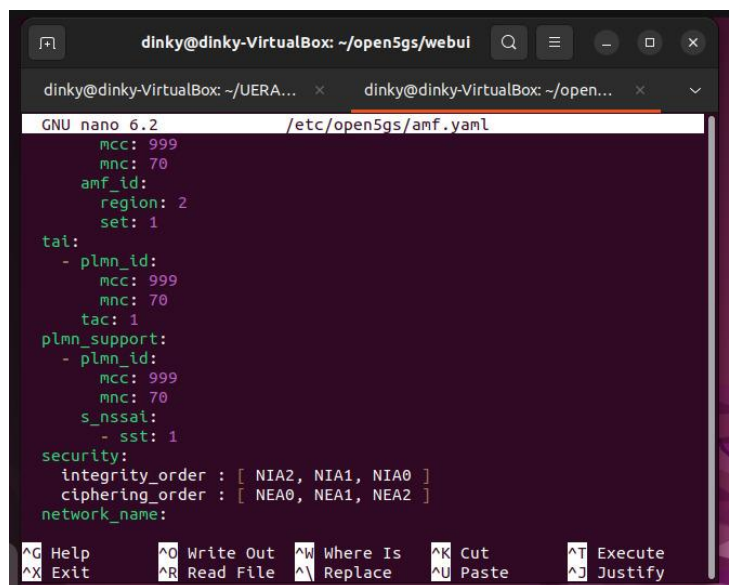
Core Configuration Files Modified:

- AMF Core Daemon Configuration (*/etc/open5gs-amfd / amf.yaml*): Updated *s_nssai* under *plmn_support* to include the target slice ($SST = 1, SD = 0x000002$).
- gNodeB Radio Configuration (*config/open5gs-gnb.yaml*): Updated supported *slices* list to advertise support for $SST = 1, SD = 2$ over the air.
- UE Device Profile Configuration (*config/open5gs-ue.yaml*): Updated *sessions*, *configured-nssai*, and *default-nssai* to request $SST = 1, SD = 2$ during network attach.
- Subscriber Database (MongoDB via WebUI): Provisioned the subscriber profile with matching authorized slice parameters ($SST = 1, SD = 000002$).

Methodology:

Step 1: Core Network Slice Configuration (AMF)

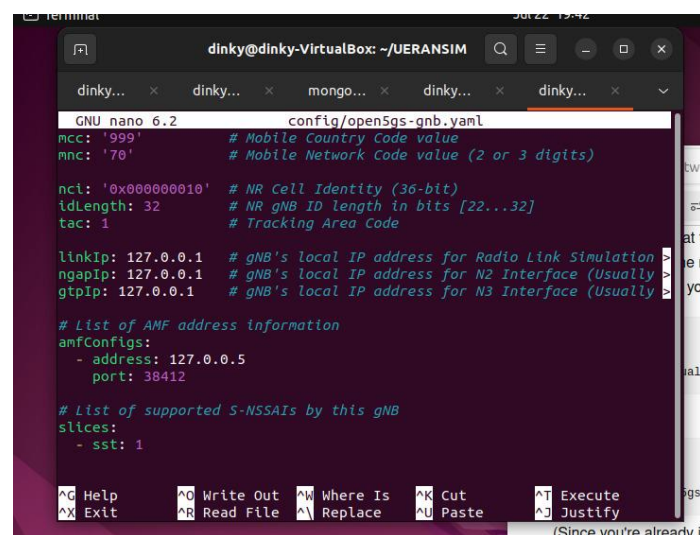
1. Open a terminal and edit the Open5GS AMF configuration file using *sudo nano /etc/open5gs/amf.yaml*
2. Under the *plmn_support* block, update *s_nssai* to support the slice differentiator *sd: 0x000002*.
3. Restart the AMF daemon using *sudo systemctl restart open5gs-amfd* and verify it is running cleanly.



```
GNU nano 6.2 /etc/open5gs/amf.yaml
mcc: 999
mnc: 70
amf_id:
  region: 2
  set: 1
tai:
- plmn_id:
  mcc: 999
  mnc: 70
  tac: 1
plmn_support:
- plmn_id:
  mcc: 999
  mnc: 70
  s_nssai:
  - sst: 1
security:
  integrity_order : [ NIA2, NIA1, NIA0 ]
  ciphering_order : [ NEA0, NEA1, NEA2 ]
network_name:
```

Step 2: gNodeB Base Station Slice Configuration

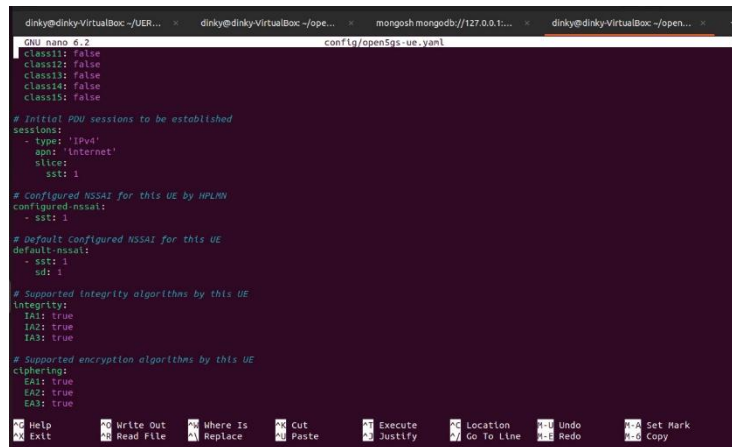
1. Navigate to the *UERANSIM* directory. Open the gNB configuration template using *nano config/open5gs-gnb.yaml*.
2. Under the *amfConfigs* and *slices* section, add the target slice details (*sst: 1*, *sd: 2*) so the virtual base station advertises availability for this slice.



```
GNU nano 6.2 config/open5gs-gnb.yaml
mcc: '999' # Mobile Country Code value
mnc: '70' # Mobile Network Code value (2 or 3 digits)
nci: '0x000000010' # NR Cell Identity (36-bit)
idlength: 32 # NR gNB ID length in bits [22..32]
tac: 1 # Tracking Area Code
linkIp: 127.0.0.1 # gNB's local IP address for Radio Link Simulation
ngapIp: 127.0.0.1 # gNB's local IP address for N2 Interface (Usually
gtip: 127.0.0.1 # gNB's local IP address for N3 Interface (Usually
# List of AMF address information
amfConfigs:
- address: 127.0.0.5
  port: 38412
# List of supported S-NSSAIs by this gNB
slices:
- sst: 1
```

Step 3: User Equipment (UE) Requested Slice Configuration

1. Open the UE configuration file using `nano config/open5gs-ue.yaml`.
2. Under the `sessions`, `configured-nssai`, and `default-nssai` sections, explicitly set `sst: 1` and `sd: 2`.
3. Save the file to ensure the UE sends this specific S-NSSAI in its NAS Registration Request.



```
dinky@dinky-VirtualBox -/UER...  dinky@dinky-VirtualBox -/ope...  mongosh mongodb://127.0.0.1...  dinky@dinky-VirtualBox -/open...
GNU nano 2.9.3 config/open5gs-ue.yaml
classid1: false
classid2: false
classid3: false
classid4: false
classid5: false

# Initial PDU sessions to be established
sessions:
- type: 'IPv4'
  apn: 'Internet'
  slice:
    sst: 1

# Configured NSSAI for this UE by HPLMN
configured-nssai:
- sst: 1

# Default Configured NSSAI for this UE
default-nssai:
- sst: 1
  sd: 1

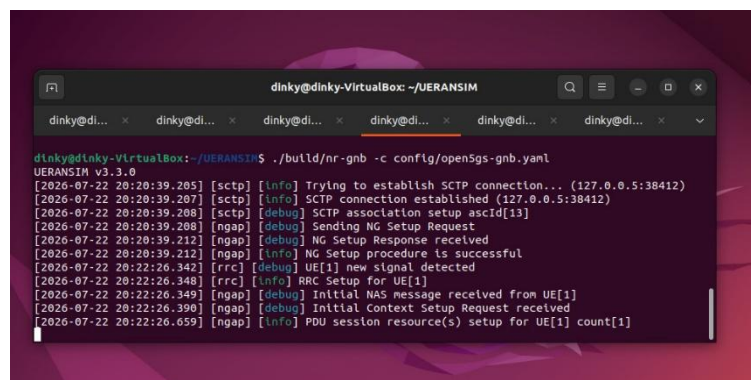
# Supported integrity algorithms by this UE
integrity:
  IA1: true
  IA2: true
  IA3: true

# Supported encryption algorithms by this UE
ciphering:
  EA1: true
  EA2: true
  EA3: true

Help  Write Out  Where Is  Cut  Paste  Execute  Location  Undo  Set Mark
Exit  Read File  Replace  Faste  Justify  Go To Line  Redo  Copy
```

Step 4: Provisioning Slice Information in MongoDB (WebUI)

1. Access the Open5GS WebUI in a browser at `localhost:9999`.
2. Open the profile for subscriber `9997000000000001`.
3. Under Slice Configurations, set SST to 1 and SD to `000002`.
4. Confirm **DNN/APN** is set to `"internet"` and save the profile to update MongoDB.



```
dinky@dinky-VirtualBox: ~/UERANSIM
dinky@di...  dinky@di...  dinky@di...  dinky@di...  dinky@di...  dinky@di...
dinky@dinky-VirtualBox: ~/UERANSIM$ ./build/nr-gnb -c config/open5gs-gnb.yaml
UERANSIM v3.3.0
[2026-07-22 20:20:39.205] [sctp] [Info] Trying to establish SCTP connection... (127.0.0.5:38412)
[2026-07-22 20:20:39.207] [sctp] [Info] SCTP connection established (127.0.0.5:38412)
[2026-07-22 20:20:39.208] [sctp] [debug] SCTP association setup ascId[13]
[2026-07-22 20:20:39.208] [ngap] [debug] Sending NG Setup Request
[2026-07-22 20:20:39.212] [ngap] [debug] NG Setup Response received
[2026-07-22 20:20:39.212] [ngap] [Info] NG Setup procedure is successful
[2026-07-22 20:22:26.342] [rrc] [debug] UE[1] new signal detected
[2026-07-22 20:22:26.348] [rrc] [Info] RRC Setup for UE[1]
[2026-07-22 20:22:26.349] [ngap] [debug] Initial NAS message received from UE[1]
[2026-07-22 20:22:26.390] [ngap] [debug] Initial Context Setup Request received
[2026-07-22 20:22:26.659] [ngap] [Info] PDU session resource(s) setup for UE[1] count[1]
```

Step 5: Network Attachment & Execution

1. Launch the virtual gNodeB:
`./build/nr-gnb -c config/open5gs-gnb.yaml`
2. In a separate terminal, launch the simulated UE:
`sudo ./build/nr-ue -c config/open5gs-ue.yaml`

- Observe the log window to confirm registration acceptance, slice authorization, and tunnel creation.

```
dinky@dinky-VirtualBox: /open5gs/webui$ sudo nano /etc/open5gs/amf.yaml
dinky@dinky-VirtualBox: /open5gs/webui$ sudo systemctl restart open5gs-amfd
dinky@dinky-VirtualBox: /open5gs/webui$ sudo systemctl status open5gs-amfd
● open5gs-amfd.service - Open5GS AMF Daemon
   Loaded: loaded (/lib/systemd/system/open5gs-amfd.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-07-22 19:04:59 IST; 13s ago
     Main PID: 5415 (open5gs-amfd)
       Tasks: 2 (limit: 4598)
      Memory: 7.5M
         CPU: 12ms
    CGroup: /system.slice/open5gs-amfd.service
            └─5415 /usr/bin/open5gs-amfd -c /etc/open5gs/amf.yaml

Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.1
[Lines 1-20/20 (END)...skipping...]
● open5gs-amfd.service - Open5GS AMF Daemon
   Loaded: loaded (/lib/systemd/system/open5gs-amfd.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-07-22 19:04:59 IST; 13s ago
     Main PID: 5415 (open5gs-amfd)
       Tasks: 2 (limit: 4598)
      Memory: 7.5M
         CPU: 12ms
    CGroup: /system.slice/open5gs-amfd.service
            └─5415 /usr/bin/open5gs-amfd -c /etc/open5gs/amf.yaml

Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.101: [sbl] INFO: Setup NF EndPoint(addr) [127.0.0.4:7777] (.../lib
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.101: [sbl] INFO: [bdb82888-85cc-41f1-88cd-15545249f77d] (NF) b-d
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.101: [sbl] INFO: [PCF] NFInstance associated [bdb82888-85cc-41f1
Jul 22 19:04:59 dinky-VirtualBox open5gs-amfd[5415]: 07/22 19:04:59.101: [sbl] INFO: Setup NF EndPoint(addr) [127.0.0.13:801] (.../lib
```

Experimental Results:

The control plane signaling, configuration modifications, and user-plane session setups were successfully executed. Below is a summary of the system validation checks:

Component / Layer	Configuration / Output Command	Operational Result
AMF Configuration	<i>/etc/open5gs/amf.yaml</i>	S-NSSAI sst: 1, sd: 0x000002 added to plmn_support
gNB Configuration	<i>config/open5gs-gnb.yaml</i>	Slice sst: 1, sd: 2 enabled on radio interface.
UE Configuration	<i>config/open5gs-ue.yaml</i>	NAS profile configured to request sst: 1, sd: 2
Subscriber Database	Open5GS WebUI (MongoDB)	Subscriber 9997000000000001 authorized for SST: 1, SD: 000002
Control Signaling	Registration accept received	AMF successfully verified requested slice against MongoDB.
User Plane Tunnel	<i>TUN interface [uesimtun0] active</i>	Dedicated PDU session established for the requested slice.

Analysis:

- Configuration Consistency:** 5G Network Slicing requires strict alignment across the entire chain. The AMF (*amf.yaml*), the base station (*open5gs-gnb.yaml*), the device

(*open5gs-ue.yaml*), and the subscriber database (MongoDB) were all explicitly configured to match $SST = 1, SD = 000002$.

- **Slice Authorization Handshake:** When the UERANSIM UE initiated attachment, its NAS registration payload contained the requested S-NSSAI. The Open5GS AMF received this request, cross-referenced it with the newly provisioned MongoDB subscriber profile, and verified that the subscriber was allowed to access $SD = 000002$.
- **PDU Session Activation:** Upon authorization, the AMF accepted the registration (Registration accept received). The Session Management Function (SMF) then established a slice-dedicated PDU session, bringing up the virtual tunnel interface (*uesimtun0*) for active user-plane traffic.

Conclusion:

This assignment successfully demonstrated the configuration, provisioning, and validation of 5G Network Slicing using Open5GS and UERANSIM. By explicitly editing the core network, radio, and device YAML configuration files and synchronizing them with MongoDB slice authorizations, the setup confirmed that subscriber slice permissions are properly enforced at the 5G Core control plane. The successful attachment and creation of the *uesimtun0* network interface confirm the correct operational end-to-end flow of a 5G Standalone (SA) network slicing architecture.